

A Behavioral-Integrity-Aware Multimodal Cognitive Readiness Framework for Adaptive Assessment

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Abstract—Most adaptive e-learning systems select assessment difficulty from a single signal: prior accuracy. This short paper presents Multimodal Cognitive Readiness Framework (MCRF), which estimates a learner’s *Cognitive Readiness Score (CSR)* by combining recent performance, real-time webcam-derived attention, response-timing integrity (to discount likely guesses), short-term performance trend, and a content-complexity estimate derived from recently consumed learning material. CSR feeds a Multimodal Closed-Loop Difficulty Engine (MCL-DE) that selects the next assessment’s difficulty and that updates as new behavioral evidence arrives. We describe the framework design, a working prototype (FastAPI backend, Next.js frontend, MediaPipe-based attention sensing, Faster-Whisper transcription, FLAN-T5 question generation) and report preliminary qualitative observations from internal prototype testing. We position this work against existing adaptive-testing and engagement-sensing literature and identify content-complexity fusion and a controlled learner study as the immediate next steps. The framework is implemented as a functional prototype and provides a foundation for future controlled educational evaluation.

Index Terms—adaptive assessment, learning analytics, cognitive readiness, attention monitoring, behavioral integrity, intelligent tutoring systems

I. INTRODUCTION

Adaptive assessment systems typically condition difficulty selection on accuracy alone [1], [2]. This is a weak proxy for readiness: a learner may answer correctly while guessing quickly or incorrectly while attentive and reasoning carefully. Separately, a body of work uses ordinary webcams and facial marker models to estimate student engagement [5]–[8], but these systems largely stop at producing a dashboard score rather than feeding an instructional decision. A third line of work generates assessment items automatically from text using transformer language models [9], [10], grounded in transcripts produced by automatic speech recognition [11], [12].

This paper integrates these threads into a single decision variable. We propose the **Cognitive Readiness Score (CSR)**, which fuses performance, attention, response-timing integrity, and short-term trend, with content complexity incorporated as an additional content-aware signal. CSR drives the **Multimodal Closed-Loop Difficulty Engine (MCL-DE)**, which selects assessment difficulty and recalibrates after every session. Unlike accuracy-only adaptive testing, MCL-DE explicitly discounts performance evidence collected under low attention or under timing patterns consistent with guessing. Unlike existing adaptive learning approaches, the proposed framework combines behavioral, performance, and content-

aware indicators within a unified cognitive readiness model for adaptive assessment.

Contributions

The contributions of this work are summarized as follows:

- 1) We introduce the Multimodal Cognitive Readiness Framework (MCRF), a closed-loop adaptive learning framework integrating behavioral, performance, and content-aware learner-state indicators.
- 2) We introduce the Cognitive Readiness Score (CSR), a behavioral-integrity-aware learner-state model that integrates performance, attention, response integrity, learning trend, and content complexity within a unified adaptive decision framework.
- 3) We develop the Multimodal Closed-Loop Difficulty Engine (MCL-DE), which continuously adapts assessment difficulty based on CSR.
- 4) We present a prototype implementation demonstrating the feasibility of real-time learner-state modeling and adaptive assessment delivery.

II. RELATED WORK

Systematic reviews of adaptive e-learning identify content-, learner-profile-, pedagogy-, and structure-level adaptation as the dominant design models [1], with AI/ML-driven personalization shown across dozens of studies to improve engagement and scores [2], [3]. More technically ambitious systems pair reinforcement learning with multi-style content adaptation [4], but adaptation is still driven by post-hoc performance rather than concurrent behavioral signals.

A parallel literature uses MediaPipe FaceMesh to derive eye-contact, head-pose, and blink-rate features for engagement classification [5], [6], with graph-based temporal models improving classification further [7], and comparative work confirming MediaPipe’s practicality against alternatives such as OpenFace [8]. In nearly every case, the engagement estimate is the end product surfaced to an instructor, not an input consumed by a downstream adaptive decision.

Automatic question generation has moved from rule-based pipelines to fine-tuned T5-family transformers, producing question sets rated as educationally relevant in a majority of cases [9], though a systematic review notes persistent weaknesses in distractor quality [10]. Whisper-family ASR models provide a practical, low-cost substitute for manual lecture captioning [11], [12], enabling transcript-grounded question generation in principle.

Gamification research reports generally positive but inconsistent effects of XP, badges, and leaderboards on motivation [13], [14], with evidence that rewards detached from the actual learning signal underperform rewards tied to genuine progress [13]. None of the systems above combine performance, attention, response-timing integrity, and trend into a single fused readiness estimate that drives difficulty selection in real time; this is the gap MCRF addresses.

Research Gap

Existing adaptive learning systems primarily rely on assessment performance as a proxy for learner state. Engagement-monitoring systems estimate attention without directly influencing pedagogical decisions, while automatic question-generation systems focus on content creation rather than adaptation. Consequently, learner readiness remains undermodeled in many adaptive learning environments. MCRF addresses this gap through a unified Cognitive Readiness Score that integrates behavioral, performance, and content-aware indicators into a closed-loop adaptive assessment framework.

III. MCRF FRAMEWORK

Fig. 1 shows the overall architecture. The MCRF framework consists of a learner interaction layer, multimodal analytics services, a Cognitive Readiness Modeling layer, and an adaptive assessment layer. The current implementation utilizes a Next.js frontend, FastAPI backend, MediaPipe-based attention analytics, Faster-Whisper transcription, and FLAN-T5 question generation. The decision layer's output — a difficulty tier and a natural-language rationale — flows back to the client, closing the loop.

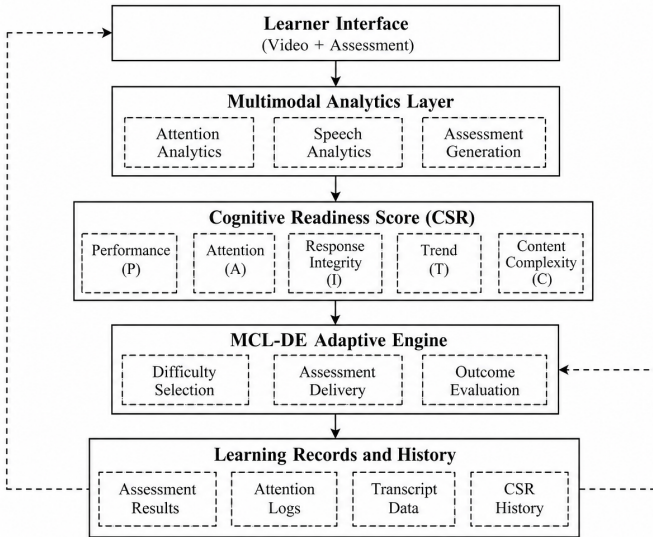


Fig. 1. MCRF architecture illustrating learner interaction, multimodal analytics, cognitive readiness estimation, adaptive difficulty selection, and closed-loop assessment feedback.

IV. COGNITIVE READINESS SCORE

CSR is defined as a weighted combination of five normalized components:

$$CSR = \alpha P + \beta A + \gamma I + \delta T + \epsilon C \quad (1)$$

where P denotes performance, A attention, I response integrity, T learning trend, and C content complexity.

$$\alpha + \beta + \gamma + \delta + \epsilon = 1 \quad (2)$$

where each term lies in $[0, 1]$ (or $[-1, 1]$ for T , rescaled). Table I defines each term.

TABLE I
CSR FEATURE DEFINITIONS

Term	Name	Definition
P	Performance	Normalized recent assessment score (0–1).
A	Attention	Webcam-derived attention score from gaze, head-pose, and blink-rate fusion (0–1).
I	Response Integrity	Function of answer time relative to the time limit; penalizes very fast (likely-guess) and very slow (likely-disengaged) responses, peaking at a thoughtful-pace interval.
T	Trend	Direction of the last 3–5 assessment scores (declining/stable/improving), encoded as $-1/0/+1$.
C	Content Complexity	Estimated linguistic/conceptual difficulty of the transcript segment just viewed (readability- or embedding-based).

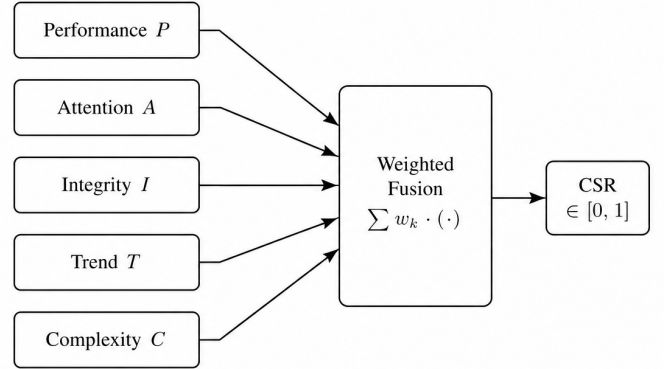


Fig. 2. CSR computation pipeline: five normalized behavioral and content signals are combined via a weighted fusion function into a single readiness score.

The integrity term is, to our knowledge, rarely formalized in webcam-based adaptive learning work; it directly addresses the validity problem of accuracy-only adaptation by discounting performance evidence obtained under timing patterns inconsistent with genuine engagement. Fig. 2 shows the computation pipeline: P , A , I , T and C are computed from data already captured by the prototype (Section V). The content-complexity component captures the linguistic and conceptual difficulty of

recently consumed learning material and contributes to CSR-based readiness estimation as a content-aware learner-state signal.

MCL-DE consumes CSR (and, currently, its four implemented constituents directly) to select a difficulty tier and to produce a short rationale string surfaced to the learner. After each assessment, observed outcomes update the trend term and, prospectively, the fusion weights, closing the loop shown in Fig. 3.

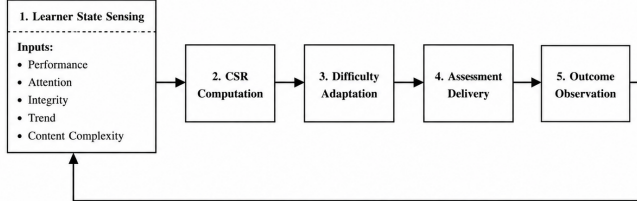


Fig. 3. MCL-DE closed loop: sensed signals drive a difficulty decision; the resulting outcome is observed and feeds the next sensing cycle.

V. PROTOTYPE IMPLEMENTATION

The MCRF prototype comprises a Next.js/React frontend and a FastAPI backend with PostgreSQL-backed persistence for student profiles, assessment sessions, learner analytics, assessment outcomes, and longitudinal learner history. The attention analytics service employs MediaPipe FaceMesh to extract eye-aspect ratio (blink rate), gaze direction, and head-pose features, which are fused into a normalized attention score from live webcam streams. The transcription service utilizes Faster-Whisper for automated speech-to-text processing, enabling transcript-driven learning analytics and assessment generation. Additional implementation considerations and future enhancements are discussed in Section VII. The question-generation service leverages FLAN-T5 to generate adaptive assessment items from transcript content, supplemented by a curated fallback question bank to ensure robustness. The Multimodal Closed-Loop Difficulty Engine (MCL-DE) integrates performance history, attention estimates, response-integrity indicators, and learning-trend signals to determine personalized assessment difficulty. Response-integrity modeling incorporates completion-time analysis to identify unusually rapid responses that may indicate low-confidence guessing. A lightweight gamification layer, including experience points (XP) and progression levels, is integrated to support learner engagement and motivation.

VI. PRELIMINARY EVALUATION

We report preliminary, qualitative observations from internal prototype testing rather than a completed learner study. Across exploratory test sessions, the attention service produced stable, plausible scores under typical webcam conditions and degraded predictably (returning a no-face state) under occlusion or poor lighting. The response-timing integrity modifier correctly flagged synthetically fast “speed-run” completions in manual testing, lowering the recommended next difficulty

TABLE II
COMPARISON WITH EXISTING ADAPTIVE LEARNING SYSTEM CLASSES
(CAPABILITY-LEVEL, NOT BENCHMARKED)

Capability	Conv. LMS	AI-Adaptive Comm.	Webcam Engmt. Re-search	MCRF
Real-time attention input	No	No	Yes	Yes
Attention drives difficulty	No	No	Rarely	Yes
Response-integrity modeling	No	No	No	Yes
Trend-aware adaptation	No	Partial	No	Yes
Content-complexity fusion	No	Partial	No	Prototype
Open, self-hostable	Varies	No	Varies	Yes

as intended. The trend modifier behaved as designed when fed manually constructed score sequences (monotonically improving, declining, and mixed). We have not yet conducted a study with independent learners, have not measured correlation between CSR and subsequent learning gain, and have not evaluated FLAN-T5 question quality against human raters at scale; these are identified explicitly, not implicitly, as ongoing work (Section VII). Table II positions the framework’s design against neighboring system classes based on architectural capability rather than empirical performance, since no class in this comparison, including ours, has been evaluated head-to-head on a shared benchmark.

VII. DISCUSSION

The proposed framework demonstrates the feasibility of integrating multimodal learner-state indicators within a unified adaptive assessment architecture. By combining behavioral and performance signals into a single Cognitive Readiness Score, the framework extends conventional accuracy-based adaptation toward a richer representation of learner readiness. The current prototype validates the end-to-end operation of the MCL-DE decision process, including attention analytics, transcript-driven assessment generation, response-integrity modeling, and adaptive difficulty selection. While the present work focuses on architectural feasibility and prototype validation, future research will investigate larger-scale learner studies, refinement of content-complexity estimation, and longitudinal evaluation of CSR-driven adaptation strategies.

VIII. CONCLUSION

This paper introduced MCRF, a multimodal adaptive assessment framework centered on Behavioral-Integrity-Aware Cognitive Readiness Modeling. The proposed Cognitive Readiness Score (CSR) integrates performance, attention, response integrity, learning trends, and content-complexity signals into a unified learner-state representation, while the Multimodal Closed-Loop Difficulty Engine (MCL-DE) uses this representation to drive adaptive assessment decisions. Unlike

conventional accuracy-based adaptive systems, MCRF explicitly considers the reliability of observed learner performance through behavioral and engagement-aware indicators. A functional prototype demonstrates the feasibility of real-time learner-state modeling, multimodal analytics, transcript-driven assessment generation, and adaptive difficulty selection within a unified architecture. Future work will focus on refining content-complexity estimation, strengthening longitudinal learner modeling, and conducting controlled evaluations against conventional score-based adaptive assessment approaches.

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